Annex A3 - ToxTemp MitoStressLUHMES assay (MSL)

1. Overview

1.1 Descriptive full-text title

Assay to test compound-derived impairment in mitochondrial respiration in human dopaminergic neurons (MitoStressLUHMES; MSL) – V1.0

1.2 Abstract

This *in vitro* test method is based on human neurons (LUHMES cells) at a stage of neurite growth. It assesses disturbances in (a) the mitochondrial respiration in general, and (b) individual mitochondrial parameters, by exposure to toxicants. The oxygen consumption rate (OCR) and the extracellular acidification rate (ECAR) are measured simultaneously using a Seahorse XF Analyzer. Basal respiration, acute response to test compound, ATP production-coupled respiration, proton leak, maximal respiration, respiratory spare capacity and non-mitochondrial respiration are determined. The data of this method are meant to predict functional impairment of mitochondria. The method has not undergone formal validation and has not been part of a ring trial. It predicts some aspects of neurotoxicity, but not all aspects covered by an *in vivo* neurotoxicity study (TG424).

2. General information

2.1 Name of test method

MitoStressLUHMES, MSL

2.2 Version number and date of deposition

This is Version 1.0 of the protocol "Assay to test compound-derived impairment in mitochondrial respiration in human dopaminergic neurons (MitoStressLUHMES; MSL) – V1.0". It was assembled in September 2022.

2.3 Summary of introduced changes in comparison to previous version(s)

There are no previous versions available.

2.4 Assigned data base name

Normal text names often do not uniquely define the method. Therefore, each method should be assigned a clearly and uniquely defined data base name.

Here this is exemplified by the names generated in the EU-ToxRisk project:

UKN1a_DART_NPC_Diff_6D_02

UKN1b_DART_NPC_Diff_4D_01

UKN2a_DART_NC_Migr_24h_04

The name is assembled (in more generic terms) from the following elements:

Axa_B_C_D_E

Axa: mandatory part of the identifier allowing unambiguous identification

A: Abbreviation / acronym of the partner depositing the assay

x: Consecutive number (referring to the partner's assay number)

a: Sub-specifier (for variants, i.e. very similar assays, but e.g. different readout or medium); not mandatory, but 'Axa' must be specific (i.e. clearly identifying) for each assay variant

B: Indication of the main intended use (max 5 letters), e.g. DART, Neuro, Liver, Lung, Renal, Redox, Stress...

C: Specifier for test system, e.g. cell type, e.g. NPC (neural precursor cells), NC (neural crest), Hep (liver cells), REN (kidney cells), PUL (lung cells) (max. 4 letters)

D: Identification of test endpoint, e.g. Diff_6D = Differentiation for 6 days; exp_24 h = exposure for 24 hours; RNA_6h = transcriptome after 6 hours; and so on (use max. 15 signs altogether; if desired in 2-3 blocks), name (and acronym) of the project partner home organisation.

E: version number.

MSL_Neuro_LUH_mito_acute

2.5 Name and acronym of the test depositor

University of Konstanz (UKN), Germany

2.6 Name and email of contact person

Prof. Dr. Marcel Leist
marcel.leist@uni-konstanz.de
Tel: +49-7531885037

2.7 Name of further persons involved

Anna-Katharina Ückert (PhD student, experimenter)
anna-katharina.ueckert@uni-konstanz.de

Mahshid Alimohammadi (PhD student, experimenter)
mahshid.alimohammadi@uni-konstanz.de

2.8 Reference to additional files of relevance

- Raw data file
- Internal SOP
- Official User Guide

3. Description of general features of the test system source

Note: *this section might be redundant with section 4. It is meant to describe the procedure of generating the cells, which are eventually used in the test method. This applies e.g. for stem cells, which have to be differentiated towards the cell type with which the method is conducted (e.g. neurons or hepatocytes derived from iPSC). See scheme for illustration.*

If this is not applicable to your test system, go directly to section 4.

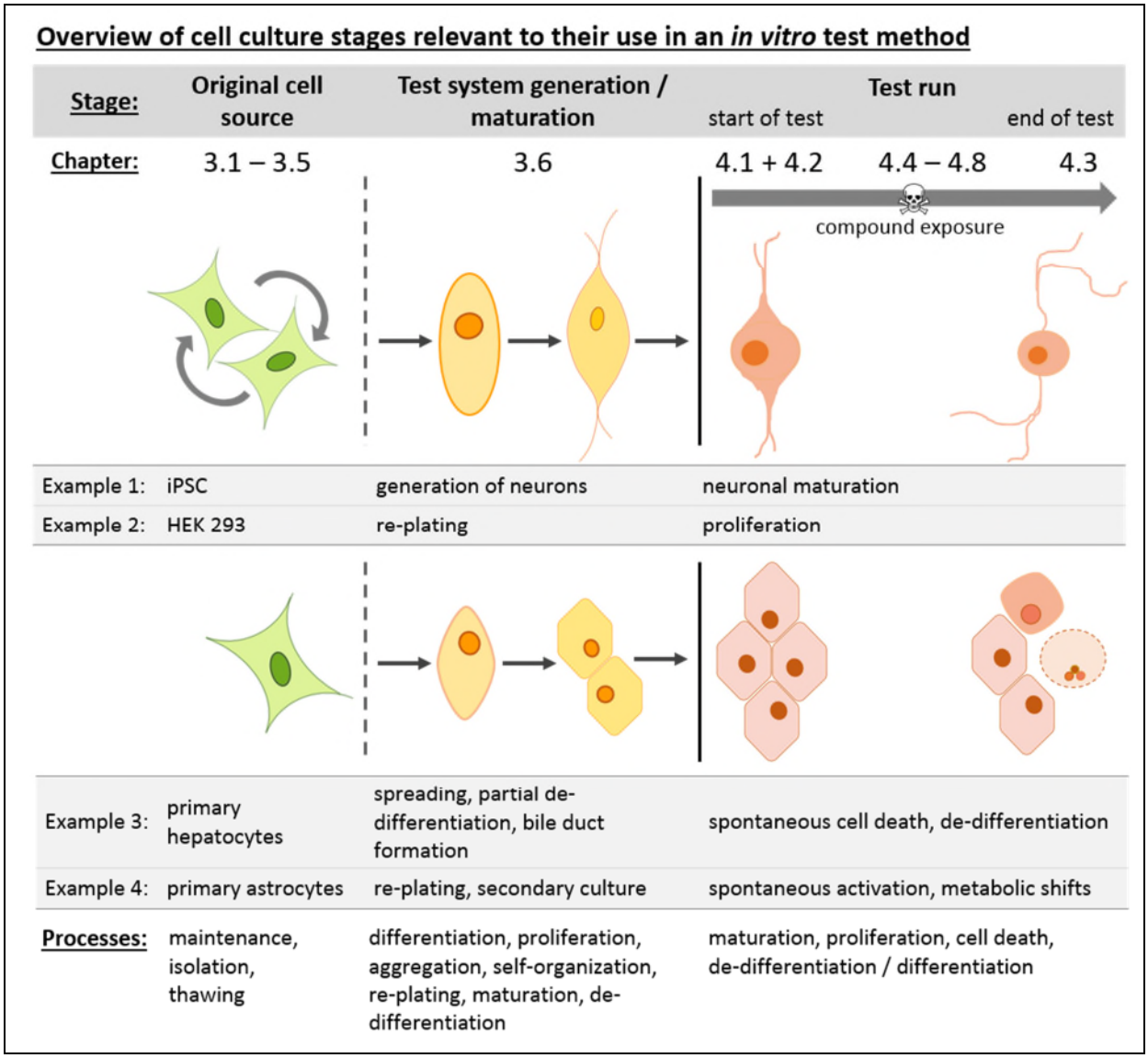


Figure 1: Overview of test system stages and where to find / deposit corresponding information in this document.

Note: Refer to overview figure to connect question numbers and cell culture stages.

3.1 Supply of source cells

The LUHMES cells have been brought from Lund (Sweden) to Konstanz in 2006 (Lotharius et al., 2005) and a masterstock has been frozen. The cells used for the test method are continuously generated by cell culture. Approximately every 4-5 years, a vial from the cell masterstock is thawed and extensively

expanded. These cells are then frozen as working stock. A new cell batch is thawed every 4 weeks from this working stock; cells are cultured until passage 20-25.

3.2 Overview of cell source component(s)

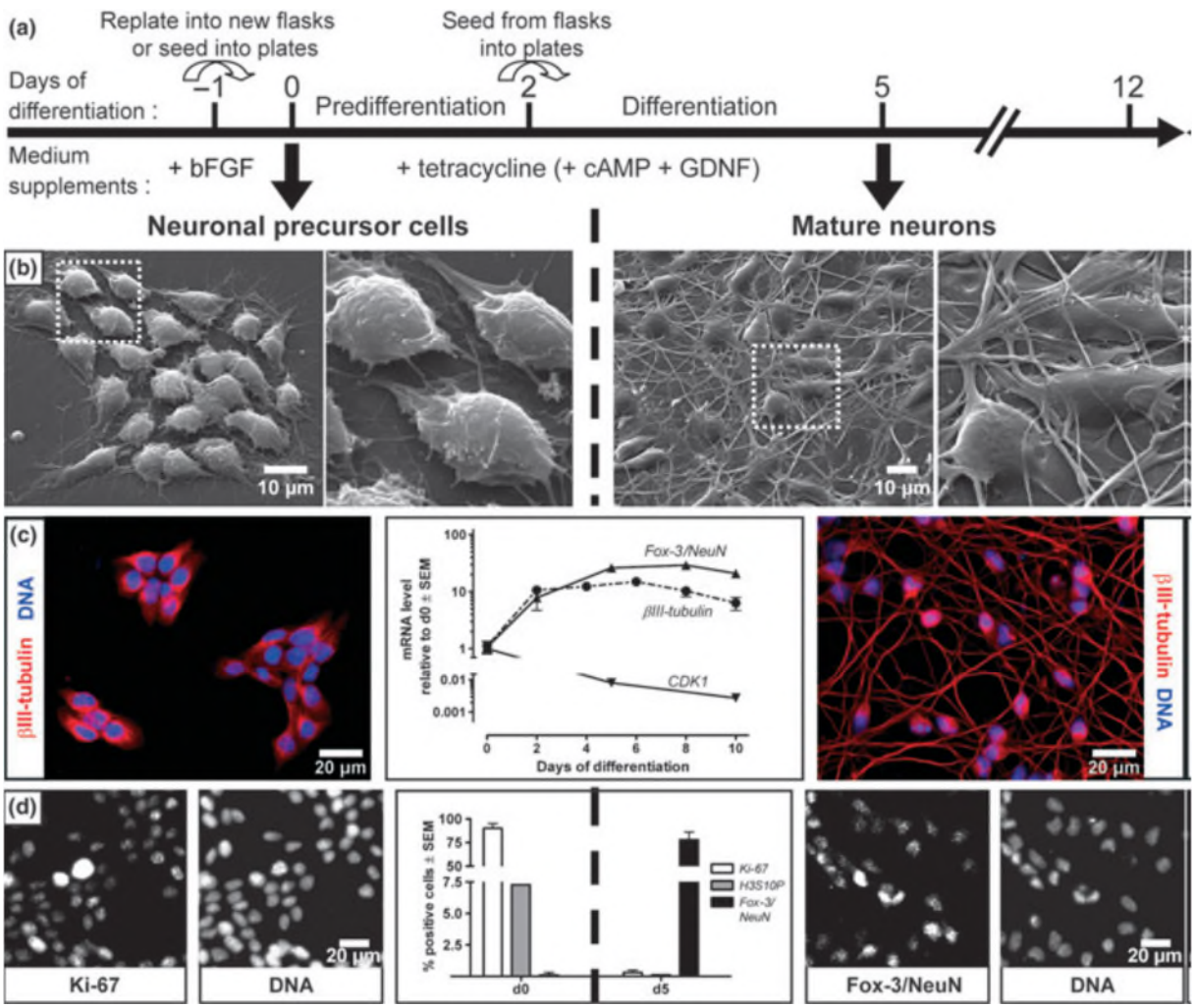
LUHMES cell line
LUHMES cells originate from the ventral mesencephalon of an 8 week old human, female fetus. They exhibit the same characteristics as MESC2.10 cells.

LUHMES cells can be differentiated into morphologically and biochemically mature **dopamine-like neurons** following exposure to tetracycline, GDNF (glial cell line-derived neurotrophic factor), and db-cAMP for 6 days.

They are usually cultured in a **2D monolayer**, but have also been shown to grow into 3D structures (Smirnova et al., 2015, Brull et al., 2020). They can be co-cultured with mouse astrocytes.

3.3 Characterization and definition of source cells

- **ATCC number**: LUHMES ATCC® CRL-2927™; LUHMES cells used at the University of Konstanz differ from LUHMES cells deposited and distributed by ATCC. The assay described in here is based on UKN (University of Konstanz) LUHMES cells.
- **Origin**: mesencephalon of an 8 week old human fetus, subclone of the tetracycline-controlled, v-myc-overexpressing human mesencephalic-derived cell line MESC2.10. MESC2.10 has been conditionally immortalized with a LINX v-myc retroviral vector with a tet-off system.
- **Gender**: female
- **Morphology**: upon differentiation, cells form neuronal network, see figure below
- **Doubling time:** approx. 14 -20 h (depending on passage)
- **Phenotype:** Dopaminergic (DA)-phenotype; tyrosine hydroxylase (TH) expression depends on the presence of cAMP in the culture medium. Under differentiation conditions, cells extend neurites after 24h. They display extensive growth cones, but synapse formation is not clear. After 6 days of differentiation, cells express voltage-dependent ion channels and show **electrical activity**.
- **Expression:** Cells express α-synuclein (Parkinson's disease) (Schildknecht et al., 2013), all standard synaptic proteins and all proteins required for AD (Alzheimer's disease) pathology (Aβ-formation, Tau hyperphosphorylation)(Scholz et al., 2013; Scholz et al., 2018).



Conversion of proliferating LUHMES cells into post-mitotic neurons. LUHMES were grown and differentiated either on glass cover slips or in multi-well plates. Cells were either fixed for microscopy or lysed for RNA extraction at different stages between day 0 and day 10 (d0–d10). **(a)** Schematic representation of the 2-step differentiation procedure, initiated by the absence of the cytokine basic fibroblast growth factor (bFGF) and addition of tetracycline. Unless mentioned otherwise, dibutyryl cAMP (cAMP) and glial cell derived neurotrophic factor (GDNF) were present throughout the differentiation. **(b)** Representative scanning electron microscopy (SEM) images of undifferentiated (d0) and differentiated (d5) LUHMES with marked squares shown at higher magnification. **(c)** LUHMES were immunostained on d0 and d5 for bIII-tubulin and nuclei were labeled by DNA staining with H-33341 dye. The mRNA expression levels of bIII-tubulin, Fox-3/NeuN and cyclin-dependent kinase 1 (CDK1) were determined after different days of maturation by RT-qPCR. **(d)** The proliferative status of d0 and d5 cells was quantified by immunostaining of Ki-67, H3S10P and Fox-3/NeuN. It is indicated as percentage of positive nuclei relative to all nuclei, as identified by DNA staining with H-33342. Quantitative data are expressed as means ± SEM from three independent differentiations.
Scholz et al., 2011

Short Tandem Repeat signature: (lab internal data)

	ATCC website	UKN LUHMES	ATCC LUHMES

Amelogenin	X	X	X
CSF1PO	13,14	13,14	13,14
D5S818	11,13	11,13	11,13
D13S317	9,11	9,11	9,11
D7S820	11,13	11,13	11,13
D16S539	11,12	11,12	11,12
vWA	14,17	14,17	14,17
THO1	7,9.3	7,9.3	7,9.3
TPOX	8	8	8
Penta D		12,13	12,13
D8S1179		12,13	12,13
FGA		19,21	19,21
D3S1358		17,18	17,18
D21S11		30,31	30,31
D18S51		12	12
Penta E		11,13	11,13

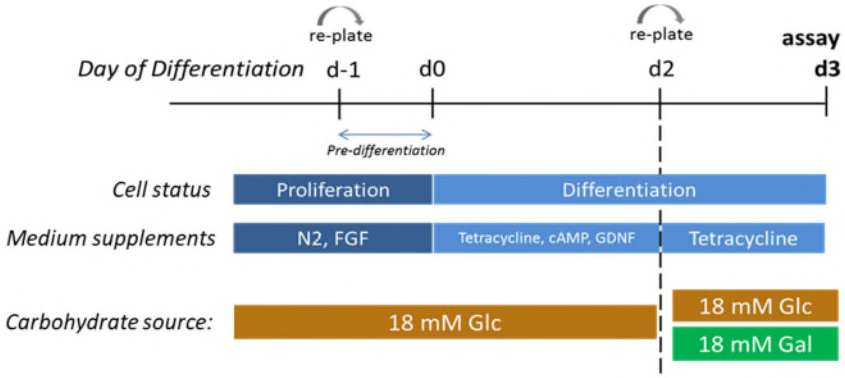
3.4 Acceptance criteria for source cell population

The cells have to be pathogen-free to be used in further experiments (regular testing for mycoplasma after thawing). Cells of the working stock are checked for marker gene expression once the stock is created, but not every thawed cell batch is checked. There only visual inspection is done. The cells should be of oval morphology, and should not form extensive aggregates and clustering. Approx. 5-10% of cell death (dead cells floating in flask) is considered normal, increased cell death one passage after thawing justifies to discard of the cell batch. The culture medium should not be yellow. Proliferating cells from working stock batches should only be used up to passage 18.

3.5 Variability and troubleshooting of source cells

- Plastic coating is critical for even cell distribution; problems with coating often leads to cell clumping and aggregation
- Cells maintained and distributed by ATCC might behave differently than cells from the stock at University of Konstanz (Gutbier et al., 2018)
- Passage number of working stock cells might influence cellular behavior
- Problems with plastic ware obtained from other suppliers than indicated have occurred in the past
- Too little or too high cell density can cause problems (confluency should never exceed 40-90%)

3.6 Differentiation towards the final test system



day -1: Proliferating LUHMES cells are seeded in proliferation medium
day 0: Medium is changed from proliferation medium to differentiation medium
day 2: Re-plating to coated Agilent Seahorse XFe24-well plates in DM without cAMP and GDNF at a density of 100,000 cells/well
day 3: Assay

Cell culture glucose media are based on Advanced DMEM/F12 supplemented with 2 mM L-glutamine. Cell culture galactose media are based on SILAC Advanced DMEM/F12 (Thermo Fisher Scientific; Cat# A2494301) supplemented with 500 µM L-lysine, 700 µM L-arginine, 18 mM galactose, 21.5 µM phonl red and 2 mM L-glutamine.
For proliferation medium, N2 (1x) and FGF (fibroblast growth factor, 40 ng/ml) are added to the desired base medium.
For differentiation medium, tetracycline (1 µg/ml, initiates differentiation by shutting down v-myc expression), GDNF (glial cell line-derived neurotrophic factor, 2 ng/ml) and db-cAMP (1 mM) are added to the desired base medium. If differentiation medium without db-cAMP and GDNF is required, the volume is replaced by the respective base medium.
Seahorse assay medium consists of Seahorse XF DMEM medium (pH 7.4) (103575-100, Agilent, Santa Clara, CA USA), supplemented with 1 mM pyruvate (792500-100G, Merck, Darmstadt, Germany), 18 mM Glc (or 18 mM Gal), 2 mM glutamine, and 2.25 µM tetracycline.

Coating of flasks and plates
Plates and flasks are coated with a poly-L-orthinine (PLO, 43 µg/ml) and fibronectin (1 µg/ml) solution in ion-exchanged and purified MilliQ-H_2O overnight in the cell culture incubator. The next day the coated plastic ware is washed with MilliQ water once and dried under sterile conditions in the cell culture hood. The coated plastic can be stored at 4°C for up to 4 weeks.

3.7 Reference / link to maintenance culture protocol

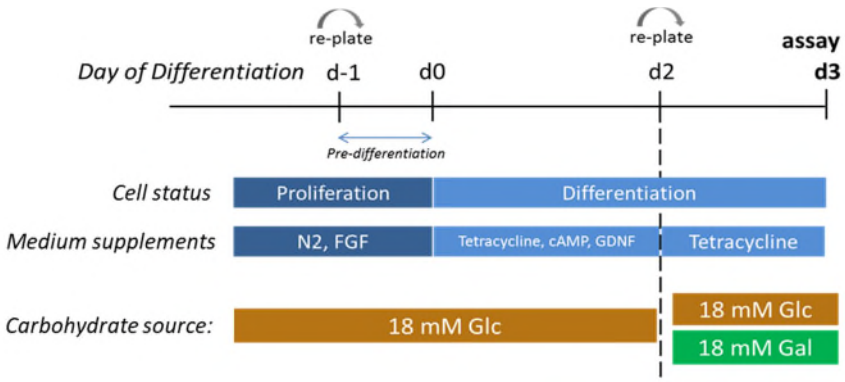
External document is available

Mainentance
Cells are maintained in proliferation medium (section 3.6) and passaged every 2-3 days (Monday – Wednesday – Friday). Minimum confluence for LUHMES cells should not fall below 40%. A confluence of 85% should not be exceeded.
For splitting, cells are counted every time and seeded in the according cell numbers:
For 2 days: 3 Mio per T75 flask
6 Mio per T175 flask
For 3 days: 1 Mio per T75 flask
2.7 Mio per T175 flask

4. Definition of the test system as used in the method

Note: *this section refers to the stage of the test system, which is then used for the test method. See scheme for illustration. If you have cells that do not need prior differentiation, give their basic characteristics here. If your test system undergoes significant changes between the maintenance culture and the use for testing, please also fill in section 3.*

4.1 Principles of the culture protocol



day -1: Proliferating LUHMES cells are seeded in proliferation medium
day 0: Medium is changed from proliferation medium to differentiation medium
day 2: Cell number reaches about 20-30 Mio per T75 flask. Cells are trypsinized and re-plated onto coated Agilent Seahorse XFe24-well plates in DM without cAMP and GDNF at a density of 100,000 cells/well in 100 µl (fill to 500 µl after 1 h of attachment)
The cartridge is rehydrated in XF Calibrant and incubated at 37°C without CO_2
The XF Analyzer is switched on
day 3: Medium change to Seahorse assay medium 1 h prior to the assay and fill cartridge with test and tool compounds, incubation at 37°C without CO_2
Assessment of OCR (inject test compound, followed by tool compounds and H-33342)
After 20 min incubation, cells are imaged using a high-content microscope (Cellomics).

4.2 Acceptance criteria for assessing the test system at its start

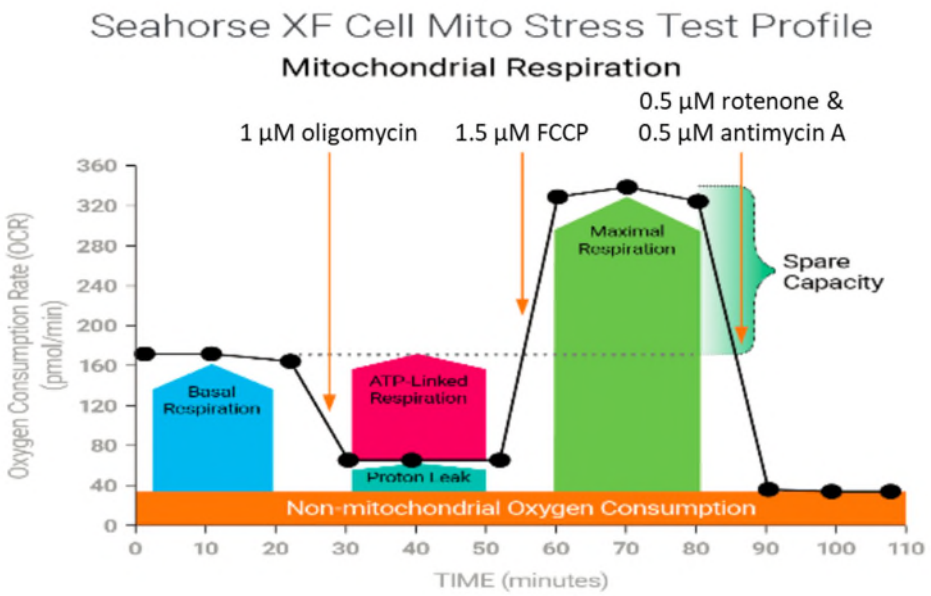
- Cells should have neurites approx. as long as their cell body when they are trypsinized for re-plating on day 2
- Cells should be evenly distributed before the medium change on day 3
- They must not be contaminated
- They should have a confluency of about 80%
- The cells should still be adherent after the medium change on day 3

→ Criteria are not quantified and are mainly based on visual inspection

4.3 Acceptance criteria for the test system at the end of compound exposure

After compound treatment, the negative controls should fulfil the following:

- OCR must not change significantly in response to control injection
- Tool compound injection should yield the characteristic MitoStress Test profile



All conditions should fulfil the following:

- Nuclei should show a healthy morphology
- Cells should still be adherent
- Oxygen levels should start at approx150 mmHg for each measurement
- Oxygen levels must not drop below 60 mmHg

→ Some criteria are not quantified and are mainly based on visual inspection

4.4 Variability of the test system and troubleshooting

<u>Causes of variability:</u>

- Problems with coating (different PLO/fibronectin batches; problems with water quality):
  → If problems occur: Wash twice after coating
  Wash with PBS instead of MilliQ-H_2O
  Don't store plates and flasks in the fridge, use them immediately
- Different cell passages:
  → Cells have different morphology and behavior the older they get; thawing a new batch might be useful
- Lots of different plates/flasks:
  → Plastic might be different, if the manufacturer delivers from a different/new lot
- Differences between the vials of one cell "lot"
- Different lots of medium and supplements
- Cells too confluent
  → impaired metabolism
  → too slow differentiation (autocrine proliferation stimulation)
  → discard cells in that case

Cell lot = cells that have all been frozen at the same time. Usually 10-15 flasks (T175) are frozen in numerous vials.
Cell batch = vial with approx. 3 Mio cells frozen
Cell passage = cells of each thawed vial are passaged up to passage 20

4.5 Metabolic capacity of the test system

Dopamine transporter is expressed and used for MPP^+ transport.

4.6 Omics characterization of the test system

This characterization was performed for the UKN4 assay. Since the 2 assays use similarly differentiated LUHMES cells, comparable results are expected.

Microarray analysis has been used to compare differentiated LUHMES on day 3 (end of UKN4) and day 6 to undifferentiated LUHMES cells (Data source: https://kops.uni-konstanz.de/bitstream/handle/123456789/28842/ Weng_288423.pdf;sequence=1).

Epigenetic modifiers have been extensively characterized (Weng et al., 2014).
Genes relevant to AD have been extensively characterized (Scholz et al., 2011).
Genes relevant for neuronal receptor composition have been extensively characterized (Loser et al., 2020; Loser et al., 2021). Genes triggered by mitochondrial toxicants have been identified (Delp et al., 2021).
There is a complete transcriptome data set on LUHMES differentiation from day 2 – day 10: It is deposited at the EBI data base under accession-ID S-TOXR1833 (public as of March 2022).

4.7 Features of the test system that reflect the *in vivo* tissue

- As neurons of the central nervous system they express the dopamine transporter DAT
- Cells grow axons and neurites in course of their differentiation
- They express various neuronal transporters and receptors (e.g. purinergic receptors, nicotinic acetylcholine receptors (Loser et al., 2020)
- They are electrophysiologically active and excitable, produce action potentials (Loser et al., 2020)

4.8 Commercial and intellectual property rights aspects of cells

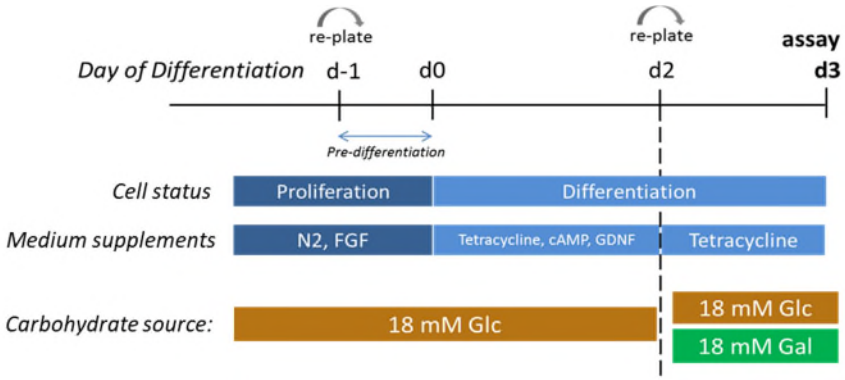
The cells are not protected by patents or any other licences.

4.9 Reference / link to the culture protocol

Chapter 3 has been answered.
Another lab-internal SOP is also available upon request to the Leist-lab.

5. Test method exposure scheme and endpoints

5.1 Exposure scheme for toxicity testing



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The cartridge is rehydrated in XF Calibrant and incubated at 37°C without CO_2
The XF Analyzer is switched on
day 3: Medium change to Seahorse assay medium 1 h prior to the assay and fill cartridge with test and tool compounds, incubation at 37°C without CO_2
Assessment of OCR (inject test compound, followed by tool compounds and H-33342)
After 20 min incubation, cells are imaged using a high-content microscope (Cellomics).

5.2 Endpoint(s) of the test method

Test endpoints: 1) Total cell count per well
2) Acute response to the test compound
3) Basal respiration
4) ATP production-coupled respiration
5) Proton leak
6) Maximal respiration
7) Respiratory spare capacity
8) Non-mitochondrial respiration

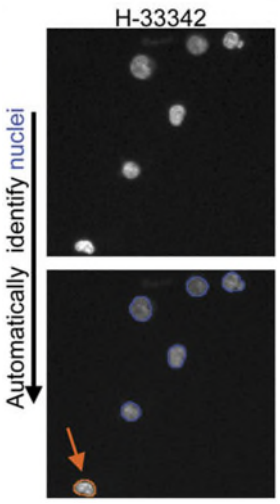
5.3 Overview of analytical method(s) to assess test endpoint(s)

1) Total cell count per well

Cells are stained with Hoechst H-33342 to allow the identification of cells. Hoechst is injected with the last tool compound and incubated for a total of 45 min in the Seahorse XF analyser (25 min during measurement, 20 min after experiment completion)

The cell staining is imaged in a Cellomics Array Scan VTI HCS reader in channel 1 (UV-Hoechst). Exposure times are set manually.

Nuclei are identified based on size, area, shape and intensity. They are counted in 25 fields and the total cell count per well is calculated based on absolute field and well size.



To calculate endpoints 2)-8), all OCR values are first normalised to the total cell count per well. For all further calculations, the first measurement overall and the first measurements after each injection are used.

2) Acute response to the test compound
   Acute response = OCR(test compound) – OCR(basal)
3) Basal respiration
   Basal respiration = OCR(test compound) – OCR(Rotenone&antimycin A)
4) ATP production-coupled respiration
   ATP production-coupled respiration = OCR(test compound) – OCR(Oligomycin)
5) Proton leak
   Proton leak = OCR(Oligomycin) – OCR(Rotenone&antimycin A)
6) Maximal respiration
   Maximal respiration = OCR(FCCP) – OCR(Rotenone&antimycin A)
7) Respiratory spare capacity
   Respiratory spare capacity = OCR(FCCP) – OCR(test compound)
8) Non-mitochondrial respiration
   Non-mitochondrial respiration = OCR(Rotenone&antimycin A)

5.4 Technical details (of e.g. endpoint measurements)

n.d.

5.5 Endpoint-specific controls / mechanistic control compounds (MCC)

n.d.

5.6 Positive controls

n.d.

5.7 Negative and unspecific controls

Negative control: solvent (0.1% DMSO final concentration).

5.8 Features relevant for cytotoxicity testing

Cells are highly sensitive to toxicants (Tong ZB 2016). Cell death is easily quantified, LDH release always shows very high baseline activity.

5.9 Acceptance criteria for the test method

n.d.

5.10 Throughput estimate

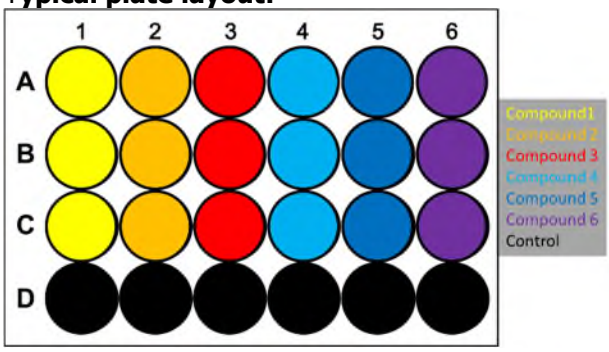
Data point = one biological replicate (→ usually 3 technical replicates); each concentration/condition of a compound counts as data point

216 data points per month

6 compounds per plate, 1 concentrations of each compound per plate (see figure) → 6 data points

3 plates can be done per day (correlates to 18 compounds → 18 data points)
→ 3 days of readout per week → 54 data points per week; 4 weeks per month → 216 data points per month.

Typical plate layout:



Typical weekly work schedule:

Monday	Tuesday	Wednesday	Thursday	Friday	Saturday	Sunday
d0 seeding		d2 seeding	d3 assay			
d-1 seeding	d0 medium change		d2 seeding	d3 assay		
				d-1 seeding	d0 medium change	
d2 seeding	d3 assay					

6. Handling details of the test method

6.1 Preparation / addition of test compounds

- Compounds are stored according to the manufacturer's instructions (e.g. 4°C, room temperature, -20°C).
- Preferable solvent is DMSO. The used DMSO is stored in a lightproof, air-tight bottle at room temperature.
- Final DMSO concentration on the cells is 0.1%
- After dissolving the compounds, which are delivered in a solid/powder form, all compound solutions are aliquoted into volumes sufficient for one experiment (i.e. one biological replicate). In this way repeated freezing and thawing and therefore damaging the compound's stability and efficiency can be avoided.

- For conducting an experiment, a compound aliquot is thawn and diluted with Seahorse assay medium to 10x final concentration with 1% DMSO.
- The dilutions are loaded into the hydrated cartridge Ports (final concentrations are given)
    a) Test compound
    b) 1 µM oligomycin
    c) 1.5 µM FCCP
    d) 0.5 µM rotenone, 0.5 µM antimycin A, 1 µg/ml H-33342

6.2 Day-to-day documentation of test execution

Plate maps are defined prior to the experiment and documented in the metadata of the experiment. Files are stored on the work group server and the online evaluation platform provided by Agilent (https://seahorseanalytics.agilent.com/Account/Login).
Concentrations and compound dilutions (if applicable) are calculated prior to the experiment.
Experimental procedures are noted manually in a paper lab book.

6.3 Practical phase of test compound exposure

The experimenter plans the experiment according to Seahorse XF Analyzer availability (has to be booked in advance) and availability of a sufficient number of cells.

The paper lab book is taken to cell culture rooms and errors are documented in there right away.

Pipetting is performed left to right, top to bottom.

Seeding errors are marked directly on the plate maps and are documented in the lab book.

6.4 Concentration settings

6 conditions can be assessed per plate. These can be used for different compounds or concentrations. Depending on potency, concentration steps can range between 1:2 and 1:10.

6.5 Uncertainties and troubleshooting

- Compound solubility in stock and during dilution is too low (stock solved in 100% DMSO, final concentration of the solvent on the cells is 0.1% DMSO)
- Some compounds show autofluorescence and interfere with the detection of H-33342.
- Focusing failure of Array Scan VTI HCS Reader (Cellomics, PA) can be a problem that produces outliers; as well as failures to save individual images.
- Highly trained/automated handling with a multistepper pipette is necessary to achieve little variance.
- Operators can get trained within 2-4 weeks. Cell seeding and medium change should be performed as fast as possible to keep cells as short as possible at room temperature. Furthermore, only 2 wells should be aspirated and refilled at once on day 3. The more practice an operator has, the faster the critical steps can be performed.
- The cartridge should not be set down harshly, since this can lead to premature injection of compounds into the calibrant solution and, therefore, a decreased final concentration on the cells

6.6 Detailed protocol (SOP)

6.7 Special instrumentation

The method requires a Seahorse XF analyser and a Cellomics Array Scan VTI HCS high content reader that may not be present in the standard lab.

6.8 Possible variations

a. further additional endpoints:
- metabolic activity (resazurin reduction)
- glutathione levels
- staining of tubulin
- analysis of differentiation markers by qPCR or immunostaining

b. other analytical endpoints:
cell viability by:
- fluorimetric measurement of resazurin conversion
- measurement of extracellular LDH
- measurement of luminescence indicating ATP content
- calcein-AM staining

c. other exposure:
- longer exposure is possible
- later exposure is possible (from day 5 on) in order to measure more mature neurons
- the medium can be changed to contain galactose instead of glucose. This increases the sensitivity of the cells to inhibitors of mitochondrial respiration (Delp et al., 2019)

6.9 Cross-reference to related test methods

UKN3a assay to test compound-derived neurite integrity impairment in human mature dopaminergic neurons after long-term compound exposure - Protocol no 202 in DB-ALM

UKN3b assay to test compound-derived neurite integrity impairment in human mature dopaminergic neurons - Protocol no 196 in DB-ALM

UKN4 (NeuriTox) assay to test compound-impaired neurite outgrowth (pre-screening)

UKN4b (MitoMet) assay to test compound-impaired neurite outgrowth with increased sensitivity to mitochondrial toxicants (pre-screening)

MCL: Seahorse measurement of compound-induced changes in the activity of the individual mitochondrial complexes in proliferating LUHMES as described in Delp et al 2019.

7. Data management

7.1 Raw data format

Seahorse XF Analyzer: Raw data and metadata are saved in a 'Wave Assay Result' file (.asyr).

Cellomics: Raw data is extracted by copy-paste in Excel files (example file available).

Cellomics normalization data is imported into the 'Wave Assay Result' file.

7.2 Outliers

1. Mathematical procedures to define outliers have not been defined. Data points that 'look' very far off are discarded. Biological outliers do practically not exist, most far data points are the result of technical problems (focus not found, only one channel imaged, etc.)
2. All raw data (incl. outliers) are stored.

a. Raw data processing to summary data

a) Cellomics normalization data
- Array Scan VTI HCS Reader (Cellomics, PA) takes images (optionally bitmap or tiff-format; 512 x 512 pixels, 8bit or 16bit)
- Images are locally analyzed using the Array Scan software, the algorithm quantifies cell count (nuclei)
- Data are copy-pasted into an Excel sheet, further analysis is done with Excel.

b) Seahorse metabolic data
- Seahorse XF Analyzer records oxygen levels and pH. OCR and ECAR are calculated directly. These readouts are saved with the meta data (.asyr format)
- For further evaluation, the Seahorse metabolic data is uploaded to Seahorse Analytics (https://seahorseanalytics.agilent.com/Account/Login)
- Cellomics normalization data is imported and applied
- Artefacts and outliers are excluded (see 4.3)
- Endpoints are automatically calculated by the pre-set "XF Cell Mito Stress Test" Assay Kit Companion View
- Data can then be extracted as Excel files or GraphPad Prism files. Alternatively, generated graphs can be saved as images (png or jpg format)

b. Curve fitting

The data are analyzed with Excel and represented with GraphPad Prism.
For the concentration curve, a nonlinear regression fit is calculated. The fitting method is least squares. If a non-linear curve fit is not possible, a linear curve fit is performed. The curve deriving from the fit is a 4-parameter log function. Treated concentrations are analyzed for deviation from control. Sometimes it is analyzed whether the deviation of neurite growth is different from the deviation of viability. This is done by two-way ANOVA + Tukey-Kramer post hoc testing. Statistics applied are one-way ANOVA (and nonparametric) with Dunnett's post test.

BMC values with their upper and lower confidence intervals (BMCU and BMCL) are calculated via the publically available online software:
http://invitrotox.uni-konstanz.de/BMCeasy/

c. Internal data storage

The data are firstly stored on the microscope computer and the Seahorse XF Analyzer computer. Then, they are exported to other servers (lab group server, university server, Agilent server), which are back-upped regularly.

d. Metadata

a) Cellomics:
The metadata are documented, stored and exported as text document (log)-files to the according scheme: (local PC)_descriptor(date and time)_XXX.log:

The following metadata are stored:

- cellinsight-pc_160429130003_AutomationControllerIni
- cellinsight-pc_160429130003_kineticprotocol
- cellinsight-pc_160429130003_protocol
- cellinsight-pc_160429130003_scan
- cellinsight-pc_160429130003_ScanIni
- cellinsight-pc_160429130003_spooling
- cellinsight-pc_160429130003.spooled

b) Seahorse: The metadata is included in the results file

e. Metadata file format

8. Prediction model and toxicological application

a. Scientific principle, test purpose and relevance

LUHMES cells used in this test method represent cells of the central nervous system with a dopaminergic phenotype.

The MSL test method assesses mitochondrial respiration. The cells are used in an early developmental stage (day 3 of differentiation) and chemical exposure occurs during this development. Therefore, the UKN4 test method assesses hazards for developmental neurotoxicity. It can be integrated into adverse outcome pathways as an important key event to predict potential adverse outcomes in humans.

b. Prediction model

n.d.

c. Prediction model setup

n.d.

d. Test performance

n.d.

e. *In vitro – in vivo* extrapolation (IVIVE)

a) The test has not been used extensively for IVIVE.
b) No special considerations known.

f. Applicability of test method

The test is sensitive to mitochondrial toxicants. E.g. cytoskeletal toxicants have no effect.

g. Incorporation in test battery

a) Strengths:
   - Automated microscopy
   - Internal controls to exclude artefacts easily

- Mechanistic information

b) Specific effects on central nervous system dopaminergic neurons.

9. Publication / validation status

a. Availability of key publications

"Progressive degeneration of human mesencephalic neuron-derived cells triggered by dopamine-dependent oxidative stress is dependent on the mixed-lineage kinase pathway."
Lotharius J, Falsig J, van Beek J, Payne S, Dringen R, Brundin P, Leist M.
J Neurosci. 2005 Jul 6;25(27):6329-42. DOI: 10.1523/JNEUROSCI.1746-05.2005. PMID: 16000623

"Rapid, complete and large-scale generation of post-mitotic neurons from the human LUHMES cell line."
Scholz D, Pöltl D, Genewsky A, Weng M, Waldmann T, Schildknecht S, Leist M.
J Neurochem. 2011 Dec;119(5):957-71. DOI: 10.1111/j.1471-4159.2011.07255.x. PMID: 21434924

"Assessment of chemical-induced impairment of human neurite outgrowth by multiparametric live cell imaging in high-density cultures."
Stiegler NV, Krug AK, Matt F, Leist M.
Toxicol Sci. 2011 May;121(1):73-87. DOI: 10.1093/toxsci/kfr034. PMID: 21342877

"Control of Abeta release from human neurons by differentiation status and RET signaling."
Scholz, D., Y. Chernyshova and M. Leist
Neurobiol Aging. 2013 Jan;34(1):184-99. DOI: 10.1016/j.neurobiolaging.2012.03.012. PMID: 22534065

"Evaluation of a human neurite growth assay as specific screen for developmental neurotoxicants."
Krug AK, Balmer NV, Matt F, Schönenberger F, Merhof D, Leist M.
Arch Toxicol. 2013 Dec;87(12):2215-31. DOI: 10.1007/s00204-013-1072-y. PMID: 23670202

"Generation of genetically-modified human differentiated cells for toxicological tests and the study of neurodegenerative diseases."
Schildknecht S, Karreman C, Pöltl D, Efrémova L, Kullmann C, Gutbier S, Krug A, Scholz D, Gerding HR, Leist M.
ALTEX. 2013;30(4):427-44. DOI: 10.14573/altex.2013.4.427. PMID: 24173167

"A LUHMES 3D dopaminergic neuronal model for neurotoxicity testing allowing long-term exposure and cellular resilience analysis."
Smirnova L, Harris G, Delp J, Valadares M, Pamies D, Hogberg HT, Waldmann T, Leist M, Hartung T.
Arch Toxicol. 2015 Dec 8. DOI: 10.1007/s00204-015-1637-z. PMID: 26647301

"Reduced Abeta secretion by human neurons under conditions of strongly increased BACE activity."
Scholz, D., Y. Chernyshova, A. K. Uckert and M. Leist
J Neurochem. 2018 Oct;147(2):256-274. DOI: 10.1111/jnc.14467. PMID: 29804308

"Major changes of cell function and toxicant sensitivity in cultured cells undergoing mild, quasi-natural genetic drift."
Gutbier S, May P, Berthelot S, Krishna A, Trefzer T, Behbehani M, Efremova L, Delp J, Gstraunthaler G, Waldmann T, Leist M.
Arch Toxicol. 2018 Dec;92(12):3487-3503. DOI: 10.1007/s00204-018-2326-5. PMID: 30298209

"A high-throughput approach to identify specific neurotoxicants/ developmental toxicants in human neuronal cell function assays."
Delp J, Gutbier S, Klima S, Hoelting L, Pinto-Gil K, Hsieh JH, Aichem M, Klein K, Schreiber F, Tice RR, Pastor M, Behl M, Leist M.

ALTEX. 2018;35(2):235-253. DOI: 10.14573/altex.1712182. Erratum in: ALTEX. 2019;36(3):505. PMID: 29423527

"Stage-specific metabolic features of differentiating neurons: Implications for toxicant sensitivity."
Delp J, Gutbier S, Cerff M, Zasada C, Niedenführ S, Zhao L, Smirnova L, Hartung T, Borlinghaus H, Schreiber F, Bergemann J, Gätgens J, Beyss M, Azzouzi S, Waldmann T, Kempa S, Nöh K, Leist M.
Toxicol Appl Pharmacol. (b) 2018 Sep 1;354:64-80. DOI: 10.1016/j.taap.2017.12.013 Erratum in: Toxicol Appl Pharmacol. 2019 Jun 1;372:70. PMID: 29278688

"Major changes of cell function and toxicant sensitivity in cultured cells undergoing mild, quasi-natural genetic drift."
Gutbier, S., P. May, S. Berthelot, A. Krishna, T. Trefzer, M. Behbehani, L. Efremova, J. Delp, G. Gstraunthaler, T. Waldmann and M. Leist
Arch Toxicol. 2018 Dec;92(12):3487-3503. DOI: 10.1007/s00204-018-2326-5. PMID: 30298209

"Development of a neurotoxicity assay that is tuned to detect mitochondrial toxicants."
Delp, J., M. Funke, F. Rudolf, A. Cediel, S. H. Bennekou, W. van der Stel, G. Carta, P. Jennings, C. Toma, I. Gardner, B. van de Water, A. Forsby and M. Leist
Arch Toxicol. 2019 Jun;93(6):1585-1608. DOI: 10.1007/s00204-019-02473-y. PMID: 31190196

"Incorporation of stem cell-derived astrocytes into neuronal organoids to allow neuro-glial interactions in toxicological studies."
Brull, M., A. S. Spreng, S. Gutbier, D. Loser, A. Krebs, M. Reich, U. Kraushaar, M. Britschgi, C. Patsch and M. Leist
ALTEX. 2020 Mar; 37(3): 409-428. DOI:10.14573/altex.1911111. PMID: 32150624

"Establishment of an a priori protocol for the implementation and interpretation of an in-vitro testing battery for the assessment of developmental neurotoxicity."
Masjosthusmann, S., J. Blum, K. Bartmann, X. Dolde, A.-K. Holzer, L.-C. Stürzl, E. H. Keßel, N. Förster, A. Dönmez, J. Klose, M. Pahl, T. Waldmann, F. Bendt, J. Kisitu, I. Suciu, U. Hübenthal, A. Mosig, M. Leist and E. Fritsche
EFSA Supporting Publications. 2020; 17(10): 1938E.
https://doi.org/10.2903/sp.efsa.2020.EN-1938

"Human neuronal signaling and communication assays to assess functional neurotoxicity."
Loser, D., J. Schaefer, T. Danker, C. Moller, M. Brull, I. Suciu, A. K. Uckert, S. Klima, M. Leist and U. Kraushaar
Arch Toxicol. 2021 Jan;95(1):229-252. DOI: 10.1007/s00204-020-02956-3. PMID: 33269408

"Neurotoxicity and underlying cellular changes of 21 mitochondrial respiratory chain inhibitors."
Delp, J., A. Cediel-Ulloa, I. Suciu, P. Kranaster, B. M. van Vugt-Lussenburg, V. Munic Kos, W. van der Stel, G. Carta, S. H. Bennekou, P. Jennings, B. van de Water, A. Forsby and M. Leist
Arch Toxicol. 2021 Feb;95(2):591-615. DOI: 10.1007/s00204-020-02970-5.PMID: 33512557

"Neurodevelopmental toxicity assessment of flame retardants using a human DNT in vitro testing battery."
Klose, J., M. Pahl, K. Bartmann, F. Bendt, J. Blum, X. Dolde, N. Forster, A. K. Holzer, U. Hubenthal, H. E. Kessel, K. Koch, S. Masjosthusmann, S. Schneider, L. C. Sturzl, S. Woeste, A. Rossi, A. Covaci, M. Behl, M. Leist, J. Tigges and E. Fritsche
Cell Biol Toxicol. 2021 May 10. DOI: 10.1007/s10565-021-09603-2. PMID: 33969458

"Functional alterations by a subgroup of neonicotinoid pesticides in human dopaminergic neurons."
Loser, D., M. G. Hinojosa, J. Blum, J. Schaefer, M. Brüll, Y. Johansson, I. Suciu, K. Grillberger, T. Danker, C. Möller, I. Gardner, G. F. Ecker, S. H. Bennekou, A. Forsby, U. Kraushaar and M. Leist
Arch Toxicol. (b) 2021 Jun;95(6):2081-2107. DOI: 10.1007/s00204-021-03031-1. PMID: 33778899

"New approach methods (NAMs) supporting read-across: Two neurotoxicity AOP-based IATA case studies."
Van der Stel, W., G. Carta, J. Eakins, J. Delp, I. Suciu, A. Forsby, A. Cediel-Ulloa, K. Attoff, F. Troger, H. Kamp, I. Gardner, B. Zdrazil, M. J. Mone, G. F. Ecker, M. Pastor, J. C. Gomez-Tamayo, A. White, E. H. J. Danen, M. Leist, P. Walker, P. Jennings, S. Hougaard Bennekou and B. Van de Water
ALTEX. 2021;38(4):615-635. DOI: 10.14573/altex.2103051. PMID: 34114044

b. (Potential) linkage to AOPs

Test method could be potentially linked to the following AOPs in AOPwiki:

- AOP 3: Inhibition of the mitochondrial complex I of nigra-striatal neurons leads to parkinsonian motor deficits.
  → Organ effects: degeneration of DA neurons of nigrostriatal pathway

c. Steps towards mechanistic validation

a) LUHMES are dopaminergic, express DAT, TH. Are of human origin, form network
b) A formal mechanistic validation has not been performed.
c) The test covers a fundamental cellular process.

d. Pre-validation or validation

No formal OECD 34 validation study has been done (eg., ring trials with a standard set of known positive and negative controls).

e. Linkage to (e.g. OECD) guidelines / regulatory use

Test is not linked to regulatory guidelines.

10. Test method transferability

a. Operator training

Experiences are required in:
- LUHMES cell culture
- multistep pipetting
- handling of Array Scan VTI HCS Reader (Cellomics, PA) and its software
- handling of XF Seahorse analyzer and its software
- Microsoft Excel
- GraphPad Prism
- Seahorse Analytics

Operator is trained and guided by a highly experienced instructor. Approximately 4 weeks will be needed for a smooth assay performance.

b. Transfer

Test system (UKN LUHMES cells) has been transferred and established to numerous other labs.

11. Safety, ethics and specific requirements

a. Specific hazards; issues of waste disposal

No specific requirements.

b. Safety data sheet (SDS)

SDS are available in the university DaMaRIS database (**Da**ngerous **Ma**terials **R**egistry **I**nformation **S**ystem).

c. Specific facilities / licenses

Work requires S1 cell culture laboratories (genetically modified cells).
No specific facilities are required.
No specific ethical approval is required.

d. Commercial aspects / intellectual property of material / procedures

To our best knowledge, no elements needed to conduct the experimental part of the test method are protected. Programs used to conduct the analysis of the data (Microsoft Excel and GraphPad Prism) need to be purchased or obtained by license agreement, however data analysis and plotting can be done with other, freely available tools.

Annex A4 - ToxTemp MitoComplexesLUHMES assay (MCL)

1. Overview

1.1 Descriptive full-text title

Assay to test compound-derived impairment in mitochondrial complex activity in human dopaminergic neurons (MitoComplexesLUHMES; MCL) – V1.0

1.2 Abstract

This *in vitro* test method is based on human neurons (LUHMES cells) at the proliferative precursor stage. Is assesses disturbances in the activity of the mitochondrial complexes (cI, cII, cIII, cIV) by exposure to toxicants. The oxygen consumption rate (OCR) is measured using a Seahorse XF Analyzer. The activity of the four individual complexes is determined. The data of this method are meant to predict (a) functional impairment of mitochondria and (b) the mechanism of action of mitochondrial toxicants. The method has not undergone formal validation and has not been part of a ring trial. It predicts some aspects of neurotoxicity, but not all aspects covered by an *in vivo* neurotoxicity study (TG424).

2. General information

2.1 Name of test method

MitoComplexesLUHMES, MCL

2.2 Version number and date of deposition

This is Version 1.0 of the protocol "Assay to test compound-derived impairment in mitochondrial complex activity in human dopaminergic neurons (MitoComplexesLUHMES; MCL) – V1.0". It was assembled in September 2022.

2.3 Summary of introduced changes in comparison to previous version(s)

There are no previous versions available.

2.4 Assigned data base name

Normal text names often do not uniquely define the method. Therefore, each method should be assigned a clearly and uniquely defined data base name.

Here this is exemplified by the names generated in the EU-ToxRisk project:

UKN1a_DART_NPC_Diff_6D_02

UKN1b_DART_NPC_Diff_4D_01

UKN2a_DART_NC_Migr_24h_04

The name is assembled (in more generic terms) from the following elements:

Axa_B_C_D_E

Axa: mandatory part of the identifier allowing unambiguous identification

A: Abbreviation / acronym of the partner depositing the assay

x: Consecutive number (referring to the partner's assay number)

a: Sub-specifier (for variants, i.e. very similar assays, but e.g. different readout or medium); not mandatory, but 'Axa' must be specific (i.e. clearly identifying) for each assay variant

B: Indication of the main intended use (max 5 letters), e.g. DART, Neuro, Liver, Lung, Renal, Redox, Stress...

C: Specifier for test system, e.g. cell type, e.g. NPC (neural precursor cells), NC (neural crest), Hep (liver cells), REN (kidney cells), PUL (lung cells) (max. 4 letters)

D: Identification of test endpoint, e.g. Diff_6D = Differentiation for 6 days; exp_24 h = exposure for 24 hours; RNA_6h = transcriptome after 6 hours; and so on (use max. 15 signs altogether; if desired in 2-3 blocks), name (and acronym) of the project partner home organisation.

E: version number.

MCL_Neuro_LUH_mito_complexes

2.5 Name and acronym of the test depositor

University of Konstanz (UKN), Germany

2.6 Name and email of contact person

Prof. Dr. Marcel Leist
marcel.leist@uni-konstanz.de
Tel: +49-7531885037

2.7 Name of further persons involved

Anna-Katharina Ückert (PhD student, experimenter)
anna-katharina.ueckert@uni-konstanz.de

Mahshid Alimohammadi (PhD student, experimenter)
mahshid.alimohammadi@uni-konstanz.de

2.8 Reference to additional files of relevance

- Raw data file
- Internal SOP
- Official User Guide

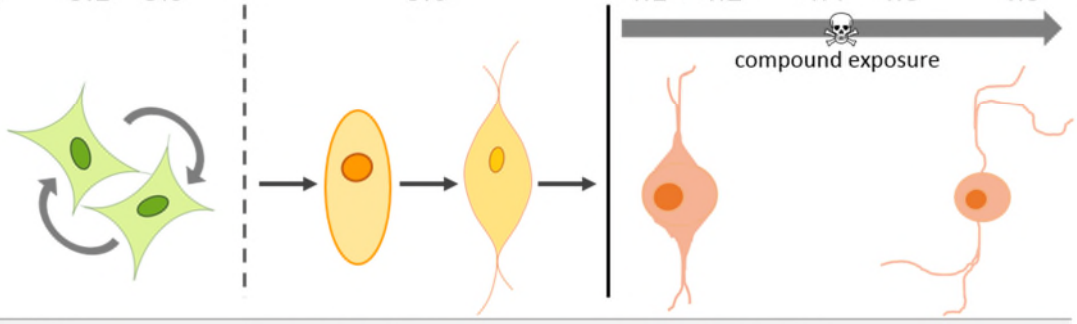
3. Description of general features of the test system source

Note: *this section might be redundant with section 4. It is meant to describe the procedure of generating the cells, which are eventually used in the test method. This applies e.g. for stem cells, which have to be differentiated towards the cell type with which the method is conducted (e.g. neurons or hepatocytes derived from iPSC). See scheme for illustration.*

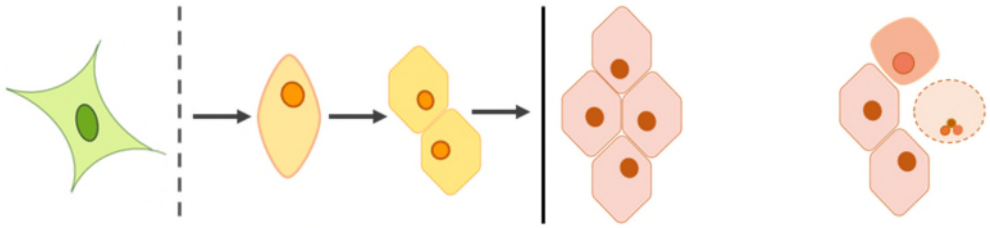
If this is not applicable to your test system, go directly to section 4.

Overview of cell culture stages relevant to their use in an *in vitro* test method

Stage:	Original cell source	Test system generation / maturation	Test run: start of test	Test run: end of test
Chapter:	3.1 – 3.5	3.6	4.1 + 4.2	4.4 – 4.8 / 4.3



Example 1:	iPSC	generation of neurons	neuronal maturation
Example 2:	HEK 293	re-plating	proliferation



Example 3:	primary hepatocytes	spreading, partial de-differentiation, bile duct formation	spontaneous cell death, de-differentiation
Example 4:	primary astrocytes	re-plating, secondary culture	spontaneous activation, metabolic shifts
Processes:	maintenance, isolation, thawing	differentiation, proliferation, aggregation, self-organization, re-plating, maturation, de-differentiation	maturation, proliferation, cell death, de-differentiation / differentiation

Overview of test system stages and where to find / deposit corresponding information in this document.

Note: Refer to overview figure to connect question numbers and cell culture stages.

3.1 Supply of source cells

The LUHMES cells have been brought from Lund (Sweden) to Konstanz in 2006 (Lotharius et al., 2005) and a masterstock has been frozen. The cells used for the test method are continuously generated by cell culture. Approximately every 4-5 years, a vial from the cell masterstock is thawed and extensively expanded. These cells are then frozen as working stock. A new cell batch is thawed every 4 weeks from this working stock; cells are cultured until passage 20-25.

3.2 Overview of cell source component(s)

LUHMES cell line

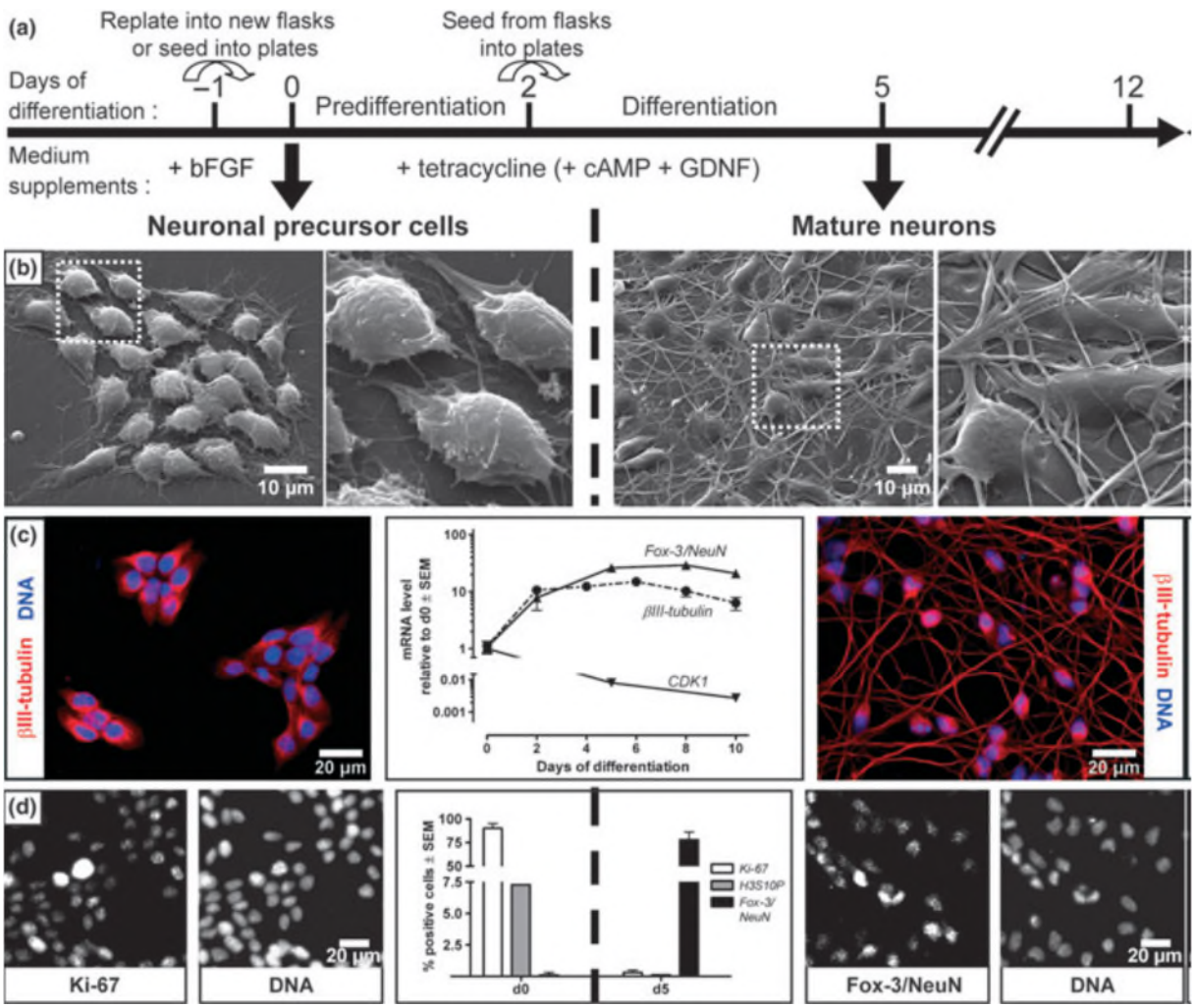
LUHMES cells originate from the ventral mesencephalon of an 8 week old human, female fetus. They exhibit the same characteristics as MESC2.10 cells.

LUHMES cells can be differentiated into morphologically and biochemically mature **dopamine-like neurons** following exposure to tetracycline, GDNF (glial cell line-derived neurotrophic factor), and db-cAMP for 6 days.

They are usually cultured in a **2D monolayer**, but have also been shown to grow into 3D structures (Smirnova et al., 2015, Brull et al., 2020). They can be co-cultured with mouse astrocytes.

3.3 Characterization and definition of source cells

- **ATCC number**: LUHMES ATCC® CRL-2927™; LUHMES cells used at the University of Konstanz differ from LUHMES cells deposited and distributed by ATCC. The assay described in here is based on UKN (University of Konstanz) LUHMES cells.
- **Origin**: mesencephalon of an 8 week old human fetus, subclone of the tetracycline-controlled, v-myc-overexpressing human mesencephalic-derived cell line MESC2.10. MESC2.10 has been conditionally immortalized with a LINX v-myc retroviral vector with a tet-off system.
- **Gender**: female
- **Morphology**: upon differentiation, cells form neuronal network, see figure below
- **Doubling time:** approx. 14 -20 h (depending on passage)
- **Phenotype:** Dopaminergic (DA)-phenotype; tyrosine hydroxylase (TH) expression depends on the presence of cAMP in the culture medium. Under differentiation conditions, cells extend neurites after 24h. They display extensive growth cones, but synapse formation is not clear. After 6 days of differentiation, cells express voltage-dependent ion channels and show **electrical activity**.
- **Expression:** Cells express α-synuclein (Parkinson's disease) (Schildknecht et al., 2013), all standard synaptic proteins and all proteins required for AD (Alzheimer's disease) pathology (Aβ-formation, Tau hyperphosphorylation)(Scholz et al., 2013; Scholz et al., 2018).



Conversion of proliferating LUHMES cells into post-mitotic neurons. LUHMES were grown and differentiated either on glass cover slips or in multi-well plates. Cells were either fixed for microscopy or lysed for RNA extraction at different stages between day 0 and day 10 (d0–d10). **(a)** Schematic representation of the 2-step differentiation procedure, initiated by the absence of the cytokine basic fibroblast growth factor (bFGF) and addition of tetracycline. Unless mentioned otherwise, dibutyryl cAMP (cAMP) and glial cell derived neurotrophic factor (GDNF) were present throughout the differentiation. **(b)** Representative scanning electron microscopy (SEM) images of undifferentiated (d0) and differentiated (d5) LUHMES with marked squares shown at higher magnification. **(c)** LUHMES were immunostained on d0 and d5 for bIII-tubulin and nuclei were labeled by DNA staining with H-33341 dye. The mRNA expression levels of bIII-tubulin, Fox-3/NeuN and cyclin-dependent kinase 1 (CDK1) were determined after different days of maturation by RT-qPCR. **(d)** The proliferative status of d0 and d5 cells was quantified by immunostaining of Ki-67, H3S10P and Fox-3/NeuN. It is indicated as percentage of positive nuclei relative to all nuclei, as identified by DNA staining with H-33342. Quantitative data are expressed as means ± SEM from three independent differentiations.
Scholz et al., 2011

Short Tandem Repeat signature: (lab internal data)

	ATCC website	UKN LUHMES	ATCC LUHMES
Amelogenin	X	X	X
CSF1PO	13,14	13,14	13,14
D5S818	11,13	11,13	11,13
D13S317	9,11	9,11	9,11
D7S820	11,13	11,13	11,13
D16S539	11,12	11,12	11,12
vWA	14,17	14,17	14,17
THO1	7,9.3	7,9.3	7,9.3
TPOX	8	8	8
Penta D		12,13	12,13
D8S1179		12,13	12,13
FGA		19,21	19,21
D3S1358		17,18	17,18
D21S11		30,31	30,31
D18S51		12	12
Penta E		11,13	11,13

3.4 Acceptance criteria for source cell population

The cells have to be pathogen-free to be used in further experiments (regular testing for mycoplasma after thawing). Cells of the working stock are checked for marker gene expression once the stock is created, but not every thawed cell batch is checked. There only visual inspection is done. The cells should be of oval morphology, and should not form extensive aggregates and clustering. Approx. 5-10% of cell death (dead cells floating in flask) is considered normal, increased cell death one passage after thawing justifies to discard of the cell batch. Culture medium should not be yellow. Proliferating cells from working stock batches should only be used up to passage 18.

3.5 Variability and troubleshooting of source cells

- Plastic coating is critical for even cell distribution; problems with coating often leads to cell clumping and aggregation
- Cells maintained and distributed by ATCC might behave differently than cells from the stock at University of Konstanz (Gutbier et al., 2018)
- Passage number of working stock cells might influence cellular behavior
- Problems with plastic ware obtained from other suppliers than indicated have occurred in the past
- Too little or too high cell density can cause problems (confluency should never exceed 40-90%)

3.6 Differentiation towards the final test system

Not applicable

3.7 Reference / link to maintenance culture protocol

External document is available

Coating of flasks and plates
plates and flasks are coated with a poly-L-orthinine (PLO, 43 µg/ml) and fibronectin (1 µg/ml) solution in ion-exchanged and purified MilliQ-H_2O overnight in the cell culture incubator. The next day the coated plastic ware is washed with MilliQ water once and dried under sterile conditions in the cell culture hood. The coated plastic can be stored at 4°C up to 4 weeks.

Proliferation medium

Advanced DMEM/F12 supplemented with 2 mM L-glutamine (G7513, Merck, Darmstadt, Germany), N2 (1x) and FGF (fibroblast growth factor, 40 ng/ml)

Mainentance
Cells are maintained in proliferation medium (section 3.6) and passaged every 2-3 days (Monday – Wednesday – Friday). Minimum confluence for LUHMES cells should not fall below 40%. A confluence of 85% should not be exceeded.
For splitting, cells are counted every time and seeded in the according cell numbers:
For 2 days: 3 Mio per T75 flask
6 Mio per T175 flask
For 3 days: 1 Mio per T75 flask
2.7 Mio per T175 flask

MAS buffer for Seahorse measurement
MAS buffer is prepared in ion-exchanged and purified MilliQ-H_2O and contains:
- 220 mM D-mannitol (M4125, Merck, Darmstadt Germany)
- 70 mM sucrose (1076511000, Merck, Darmstadt, Germany)
- 10 mM KH2PO4 (30407, Riedel-de Haën, Seelze, Germany)
- 5 mM MgCl2 (2189.1, Roth. Karlsruhe, Germany)
- 2 mM HEPES (HN77.4, Roth, Karlsruhe, Germany)
- 1 mM EGTA (E0396, Merck, Darmstadt, Germany)
- 0.4% (w/v) fatty acid free BSA (8076.5, Roth, Karlsruhe, Germany)

It is sterile filtered and stored at 4°C until use. 25 µg/ml digitonin (D141, Merck, Darmstadt, Germany) and 1 mM ADP (A5285, Merck, Darmstadt, Germany) are added right before the experiment.

4. Definition of the test system as used in the method

Note: *this section refers to the stage of the test system, which is then used for the test method. See scheme for illustration. If you have cells that do not need prior differentiation, give their basic characteristics here. If your test system undergoes significant changes between the maintenance culture and the use for testing, please also fill in section 3.*

4.1 Principles of the culture protocol

LUHMES cells are trypsinized and re-seeded into coated Agilent Seahorse XFe24-well plates in 100 µl of proliferation medium either 1 day or 2 days prior to the experiment:
- 1 day: 60'000 cells per well
- 2 days: 40'000 cells per well

After 1 h of attachment, 400 µl of proliferation medium are added to each well.

1 day prior to the assay, the cartridge is rehydrated with XF Calibrant at 37°C without CO_2 and the XF analyser is switched on.

After incubating for 1 (or 2) days as desired, the medium is replaced by MAS buffer freshly supplemented with ADP and digitonin right before the start of measurement.

4.2 Acceptance criteria for assessing the test system at its start

- Cells should have a confluency of about 80% before seeding
- Cells should be evenly distributed on the day of the assay
- They must not be contaminated

→ Criteria are not quantified and are mainly based on visual inspection

4.3 Acceptance criteria for the test system at the end of compound exposure

After compound treatment, the negative controls should fulfil the following:

- OCR must not change significantly in response to control injection

All conditions should fulfil the following:

- Oxygen levels should start at approx. 150 mmHg for each measurement
- Oxygen levels must not drop below 60 mmHg

4.4 Variability of the test system and troubleshooting

<u>Causes of variability:</u>

- Problems with coating (different PLO/fibronectin batches; problems with water quality):
  → If problems occur: Wash twice after coating
  Wash with PBS instead of MilliQ-H_2O
  Don't store plates and flasks in the fridge, use them immediately
- Different cell passages:
  → Cells have different morphology and behavior the older they get; thawing a new batch might be useful
- Lots of different plates/flasks:
  → Plastic might be different, if the manufacturer delivers from a different/new lot
- Differences between the vials of one cell "lot"
- Different lots of medium and supplements
- Cells too confluent
  → impaired metabolism
  → too slow differentiation (autocrine proliferation stimulation)
  → discard cells in that case

Cell lot = cells that have all been frozen at the same time. Usually 10-15 flasks (T175) are frozen in numerous vials.
Cell batch = vial with approx. 3 Mio cells frozen
Cell passage = cells of each vial thawn are passaged up to passage 20

4.5 Metabolic capacity of the test system

n.d.

4.6 Omics characterization of the test system

n.d.

4.7 Features of the test system that reflect the *in vivo* tissue

Mitochondrial complexes are expressed and functional.

4.8 Commercial and intellectual property rights aspects of cells

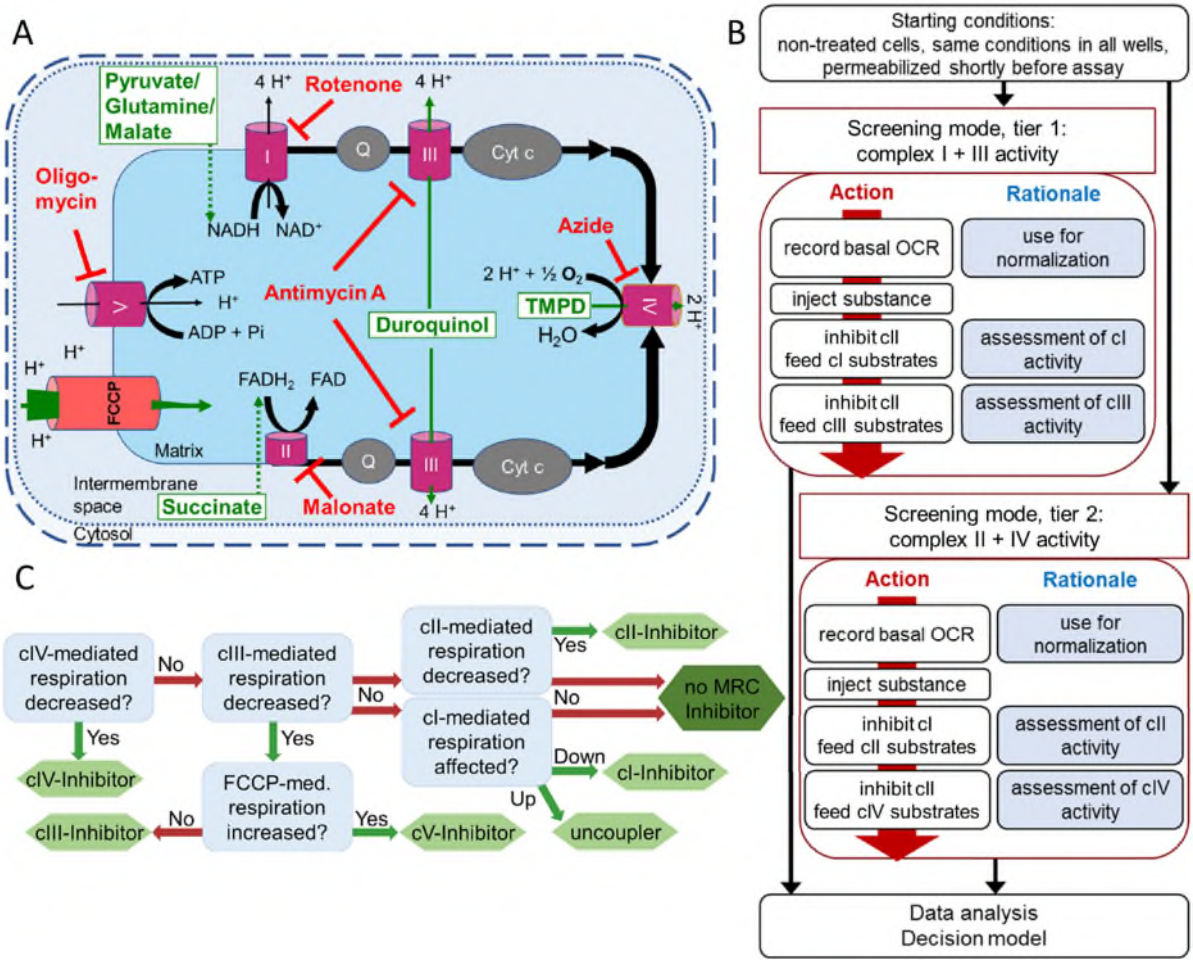
The cells are not protected by patents or any other licences.

4.9 Reference / link to the culture protocol

Chapter 3 has been answered.
Another lab-internal SOP is also available upon request to the Leist-lab.

5. Test method exposure scheme and endpoints

5.1 Exposure scheme for toxicity testing



Measurement of mitochondrial respiratory chain (MRC) complex inhibition (adapted from (Delp 2019)). (A) Specific substrates (green) and inhibitors (red) of individual MRC complexes as well as the uncoupler FCCP (orange) are shown. (B) A two-tiered screening approach was used to determine inhibitory effects of compounds on individual MRC complexes. Proliferating LUHMES cells were seeded at a density of 60,000 cells/well in proliferation medium 1 day prior to the experiment. Shortly before the assay, the cells were permeabilised by replacing the medium with MAS-buffer freshly supplemented with ADP and digitonin. The basal OCR was recorded

before the substance of interest was injected. Then, a sequence of MRC complex substrates and inhibitors was added to determine cI-cIV activity. All OCR measurements were normalized to the respective basal OCR. For each complex, the normalized OCR under treated conditions was normalized to the control to calculate the modified complex activity. In tier 1, cI substrates paired with the cII inhibitor were added first to determine cI activity. Secondly, the cIII substrate and the cI inhibitor were injected to assess cIII activity. In tier 2, the first injection consisted of the cII substrate and the cI inhibitor, it allowed the quantification of cII activity. The second injection of cIV substrates paired with the cII inhibitor permitted the quantification of cIV activity. (C) The depicted decision model was used to translate the results of the tier 1 and tier 2 screening into a classification of the substance of interest (green hexagons).

5.2 Endpoint(s) of the test method

Test endpoints: 1) cI activity
2) cII activity
3) cIII activity
4) cIV activity

5.3 Overview of analytical method(s) to assess test endpoint(s)

The individual complex activities are calculated as follows:
All OCR values are normalised to the last measurement before compound injection. Then, the first OCR measurement after selectively feeding a complex is compared to the average of the control. Complex activity is expressed in % of control.

5.4 Technical details (of e.g. endpoint measurements)

n.d.

5.5 Endpoint-specific controls / mechanistic control compounds (MCC)

cI inhibitor: 0.5 µM rotenone

cII inhibitor: 5 mM Malonate

cIII inhibitor: 0.5 µM antimycin A

cIV inhibitor: Sodium azide

uncoupler: 1.5 µM FCCP

5.6 Positive controls

n.d.

5.7 Negative and unspecific controls

Negative control: solvent (0.1% DMSO final concentration). MG132 100 nM.

5.8 Features relevant for cytotoxicity testing

Cells are highly sensitive to toxicants (Tong ZB 2016). Cell death is easily quantified, LDH release always shows very high baseline activity.

5.9 Acceptance criteria for the test method

n.d.

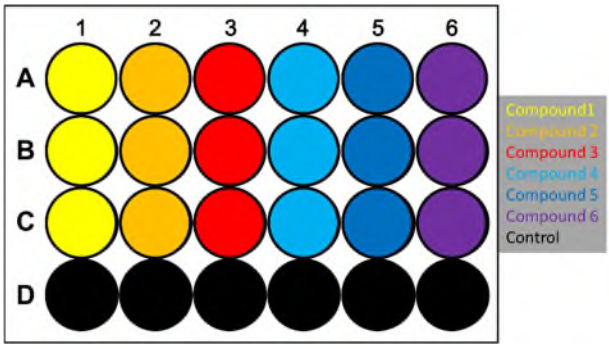
5.10 Throughput estimate

Data point = one biological replicate (→ usually 3 technical replicates); each concentration/condition of a compound counts as data point

192 data points per month

2 plates have to be run to assess the activity of all 4 complexes.
6 compounds per plate set, 1 concentrations of each compound per plate set (see figure) → 6 data points

2 plate sets can be done per day (correlates to 12 compounds → 12 data points)
→ 4 days of readout per week → 48 data points per week; 4 weeks per month → 192 data points per month.
T**ypical plate layout:**



Typical weekly work schedule:

Monday	Tuesday	Wednesday	Thursday	Friday	Saturday	Sunday
Seeding	Readout					
Seeding		Readout				
		Seeding	Readout			
		Seeding		Readout		

6. Handling details of the test method

6.1 Preparation / addition of test compounds

- Compounds are stored according to the manufacturer's instructions (e.g. 4°C, room temperature, -20°C).
- Preferable solvent is DMSO. The used DMSO is stored in a lightproof, air-tight bottle at room temperature.
- Final DMSO concentration on the cells is 0.1%

- After dissolving the compounds, which are delivered in a solid/powder form, all compound solutions are aliquoted into volumes sufficient for one experiment (i.e. one biological replicate). In this way repeated freezing and thawing and therefore damaging the compound's stability and efficiency can be avoided.
- For conducting an experiment, a compound aliquot is thawed and diluted with MAS buffer freshly supplemented with digitonin and ADP to 10x final concentration with 1% DMSO.
- The dilutions are loaded into the hydrated cartridge Ports (final concentrations are given)
    - Tier 1
        a) Test compound
        b) cI substrates: 5 mM pyruvate, 2 mM L-glutamine, 2.5 mM malate
           cII inhibitor:5 mM malonate
        c) cIII substrate: 250 µM duroquinol
           cI inhibitor: 0.5 µM rotenone
    - Tier 2
        a) Test compound
        b) cII substrate: 10 mM succinate
           cI inhibitor: 0.5 µM rotenone
        c) cIV substrates: 125 µM TMPD, 2 mM L-ascorbis acid
           cII inhibitor: 5 mM malonate

6.2 Day-to-day documentation of test execution

Plate maps are defined prior to the experiment and documented in the metadata of the experiment. Files are stored on the work group server.
Concentrations and compound dilutions (if applicable) are calculated prior to the experiment.
Experimental procedures are noted manually in a paper lab book.

6.3 Practical phase of test compound exposure

The experimenter plans the experiment according to Seahorse XF Analyzer availability (has to be booked in advance) and the availability of a sufficient number of cells.

The paper lab book is taken to cell culture rooms and errors are documented in there right away.

Pipetting is performed left to right, top to bottom.

Seeding errors are marked directly on the plate maps and are documented in the lab book.

6.4 Concentration settings

6 conditions can be assessed per plate set. These can be used for different compounds or concentrations. Depending on potency, concentration steps can range between 1:2 and 1:10.

6.5 Uncertainties and troubleshooting

- Compound solubility in stock and during dilution is too low (stock solved in 100% DMSO, the final concentration of the solvent on the cells is 0.1% DMSO)
- Highly trained/automated handling with a multistepper pipette is necessary to achieve little variance.
- Operators can get trained within 2-4 weeks. Cell seeding and medium change should be performed as fast as possible to keep cells as short as possible at room temperature. Furthermore, only 2 wells should be aspirated and refilled at once on day 3. The more practice an operator has, the faster the critical steps can be performed.

- The cartridge should not be set down harshly, since this can lead to premature injection of compounds into the calibrant solution and, therefore, a decreased final concentration on the cells

6.6 Detailed protocol (SOP)

n.d.

6.7 Special instrumentation

The method requires a Seahorse XF analyser that may not be present in the standard lab.

6.8 Possible variations

a) further additional endpoints:
n.d.

b) other analytical endpoints:
n.d.

c) other exposure:
- later exposure is possible (differentiated LUHMES)

6.9 Cross-reference to related test methods

UKN3a assay to test compound-derived neurite integrity impairment in human mature dopaminergic neurons after long-term compound exposure - Protocol no 202 in DB-ALM

UKN3b assay to test compound-derived neurite integrity impairment in human mature dopaminergic neurons - Protocol no 196 in DB-ALM

UKN4 (NeuriTox) assay to test compound-impaired neurite outgrowth (pre-screening)

UKN4b (MitoMet) assay to test compound-impaired neurite outgrowth with increased sensitivity to mitochondrial toxicants (pre-screening)

MSL: Seahorse MitoStress Test of day 3 LUHMES cells to test compound-induced changes in mitochondrial respiratory parameters (under glucose or galactose conditions)

7. Data management

7.1 Raw data format

Seahorse XF Analyzer: Raw data and metadata are saved in a 'Wave Assay Result' file (.asyr).

7.2 Outliers

1. Mathematical procedures to define outliers have not been defined. Data points that 'look' very far off are discarded. Biological outliers do practically not exist, most far data points are the result of technical problems (focus not found, only one channel imaged, etc.)
2. All raw data (incl. outliers) are stored.

7.3 Raw data processing to summary data

- Seahorse XF Analyzer records oxygen levels and pH. OCR and ECAR are calculated directly. These readouts are saved with the meta data (.asyr format)
- Artefacts and outliers are excluded (see 4.3)
- All OCR values are normalized to the last measurement before injection, respectively
- For further evaluation, the Seahorse metabolic data is exported into Excel
- The first OCR measurement after selectively feeding a complex is compared to the average of the control
- Complex activity is expressed in % of control

7.4 Curve fitting

The data are analyzed with Excel and represented with GraphPad Prism.
For the concentration curve, a nonlinear regression fit is calculated. The fitting method is least squares. If a non-linear curve fit is not possible, a linear curve fit is performed. The curve deriving from the fit is a 4-parameter log function. Treated concentrations are analyzed for deviation from control. Sometimes it is analyzed whether the deviation of neurite growth is different from the deviation of viability. This is done by two-way ANOVA + Tukey-Kramer post hoc testing. Statistics applied are one-way ANOVA (and nonparametric) with Dunnett's post test.

BMC values with their upper and lower confidence intervals (BMCU and BMCL) are calculated via the publically available online software:
http://invitrotox.uni-konstanz.de/BMCeasy/

7.5 Internal data storage

The data are firstly stored on the Seahorse XF Analyzer computer. Then, they are exported to other servers (lab group server, university server), which are back-upped regularly.

7.6 Metadata

The metadata is included in the results file

7.7 Metadata file format

8. Prediction model and toxicological application

8.1 Scientific principle, test purpose and relevance

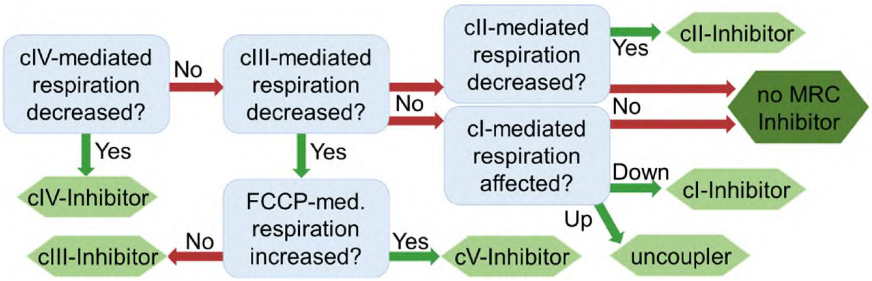
LUHMES cells used in this test method represent cells of the central nervous system with a dopaminergic phenotype.

The MCL test method assess the activity of individual mitochondrial complexes. The cells are used in proliferative precursor stage. It can be integrated into adverse outcome pathways as an important key event to predict potential adverse outcomes in humans.

8.2 Prediction model

A complex is inhibited if its activity is decreased by more than 2xSD of baseline variation. In the case of cI, uncoupling can be observed if the activity is increased by more than 2xSD of baseline variation.

Primary inhibition (mechanism of action of the compound) or secondary inhibition (due to the inhibition of an upstream complex) are decided using the following scheme.



8.3 Prediction model setup

The prediction model was published in Delp et al (2019).

8.4 Test performance

n.d.

8.5 *In vitro – in vivo* extrapolation (IVIVE)

The test has not been used extensively for IVIVE.
No special considerations known.

8.6 Applicability of test method

The test is sensitive to mitochondrial complex inhibitors. E.g. cytoskeletal toxicants have no effect.

8.7 Incorporation in test battery

Strengths:
- Internal controls to exclude artefacts easily
- Mechanistic information

9. Publication / validation status

9.1 Availability of key publications

"Progressive degeneration of human mesencephalic neuron-derived cells triggered by dopamine-dependent oxidative stress is dependent on the mixed-lineage kinase pathway."
Lotharius J, Falsig J, van Beek J, Payne S, Dringen R, Brundin P, Leist M.
J Neurosci. 2005 Jul 6;25(27):6329-42. DOI: 10.1523/JNEUROSCI.1746-05.2005. PMID: 16000623

"Rapid, complete and large-scale generation of post-mitotic neurons from the human LUHMES cell line."
Scholz D, Pöltl D, Genewsky A, Weng M, Waldmann T, Schildknecht S, Leist M.
J Neurochem. 2011 Dec;119(5):957-71. DOI: 10.1111/j.1471-4159.2011.07255.x. PMID: 21434924

"Assessment of chemical-induced impairment of human neurite outgrowth by multiparametric live cell imaging in high-density cultures."
Stiegler NV, Krug AK, Matt F, Leist M.

Toxicol Sci. 2011 May;121(1):73-87. DOI: 10.1093/toxsci/kfr034. PMID: 21342877

"Control of Abeta release from human neurons by differentiation status and RET signaling."
Scholz, D., Y. Chernyshova and M. Leist
Neurobiol Aging. 2013 Jan;34(1):184-99. DOI: 10.1016/j.neurobiolaging.2012.03.012. PMID: 22534065

"Evaluation of a human neurite growth assay as specific screen for developmental neurotoxicants."
Krug AK, Balmer NV, Matt F, Schönenberger F, Merhof D, Leist M.
Arch Toxicol. 2013 Dec;87(12):2215-31. DOI: 10.1007/s00204-013-1072-y. PMID: 23670202

"Generation of genetically-modified human differentiated cells for toxicological tests and the study of neurodegenerative diseases."
Schildknecht S, Karreman C, Pöltl D, Efrémova L, Kullmann C, Gutbier S, Krug A, Scholz D, Gerding HR, Leist M.
ALTEX. 2013;30(4):427-44. DOI: 10.14573/altex.2013.4.427. PMID: 24173167

"A LUHMES 3D dopaminergic neuronal model for neurotoxicity testing allowing long-term exposure and cellular resilience analysis."
Smirnova L, Harris G, Delp J, Valadares M, Pamies D, Hogberg HT, Waldmann T, Leist M, Hartung T.
Arch Toxicol. 2015 Dec 8. DOI: 10.1007/s00204-015-1637-z. PMID: 26647301

"Reduced Abeta secretion by human neurons under conditions of strongly increased BACE activity."
Scholz, D., Y. Chernyshova, A. K. Uckert and M. Leist
J Neurochem. 2018 Oct;147(2):256-274. DOI: 10.1111/jnc.14467. PMID: 29804308

"Major changes of cell function and toxicant sensitivity in cultured cells undergoing mild, quasi-natural genetic drift."
Gutbier S, May P, Berthelot S, Krishna A, Trefzer T, Behbehani M, Efremova L, Delp J, Gstraunthaler G, Waldmann T, Leist M.
Arch Toxicol. 2018 Dec;92(12):3487-3503. DOI: 10.1007/s00204-018-2326-5. PMID: 30298209

"A high-throughput approach to identify specific neurotoxicants/ developmental toxicants in human neuronal cell function assays."
Delp J, Gutbier S, Klima S, Hoelting L, Pinto-Gil K, Hsieh JH, Aichem M, Klein K, Schreiber F, Tice RR, Pastor M, Behl M, Leist M.
ALTEX. 2018;35(2):235-253. DOI: 10.14573/altex.1712182. Erratum in: ALTEX. 2019;36(3):505. PMID: 29423527

"Stage-specific metabolic features of differentiating neurons: Implications for toxicant sensitivity."
Delp J, Gutbier S, Cerff M, Zasada C, Niedenführ S, Zhao L, Smirnova L, Hartung T, Borlinghaus H, Schreiber F, Bergemann J, Gätgens J, Beyss M, Azzouzi S, Waldmann T, Kempa S, Nöh K, Leist M.
Toxicol Appl Pharmacol. (b) 2018 Sep 1;354:64-80. DOI: 10.1016/j.taap.2017.12.013 Erratum in: Toxicol Appl Pharmacol. 2019 Jun 1;372:70. PMID: 29278688

"Major changes of cell function and toxicant sensitivity in cultured cells undergoing mild, quasi-natural genetic drift."
Gutbier, S., P. May, S. Berthelot, A. Krishna, T. Trefzer, M. Behbehani, L. Efremova, J. Delp, G. Gstraunthaler, T. Waldmann and M. Leist
Arch Toxicol. 2018 Dec;92(12):3487-3503. DOI: 10.1007/s00204-018-2326-5. PMID: 30298209

"Development of a neurotoxicity assay that is tuned to detect mitochondrial toxicants."
Delp, J., M. Funke, F. Rudolf, A. Cediel, S. H. Bennekou, W. van der Stel, G. Carta, P. Jennings, C. Toma, I. Gardner, B. van de Water, A. Forsby and M. Leist
Arch Toxicol. 2019 Jun;93(6):1585-1608. DOI: 10.1007/s00204-019-02473-y. PMID: 31190196

"Incorporation of stem cell-derived astrocytes into neuronal organoids to allow neuro-glial interactions in toxicological studies."
Brull, M., A. S. Spreng, S. Gutbier, D. Loser, A. Krebs, M. Reich, U. Kraushaar, M. Britschgi, C. Patsch and M. Leist
ALTEX. 2020 Mar; 37(3): 409-428. DOI:10.14573/altex.1911111. PMID: 32150624

"Establishment of an a priori protocol for the implementation and interpretation of an in-vitro testing battery for the assessment of developmental neurotoxicity."
Masjosthusmann, S., J. Blum, K. Bartmann, X. Dolde, A.-K. Holzer, L.-C. Stürzl, E. H. Keßel, N. Förster, A. Dönmez, J. Klose, M. Pahl, T. Waldmann, F. Bendt, J. Kisitu, I. Suciu, U. Hübenthal, A. Mosig, M. Leist and E. Fritsche
EFSA Supporting Publications. 2020; 17(10): 1938E.
https://doi.org/10.2903/sp.efsa.2020.EN-1938

"Human neuronal signaling and communication assays to assess functional neurotoxicity."
Loser, D., J. Schaefer, T. Danker, C. Moller, M. Brull, I. Suciu, A. K. Uckert, S. Klima, M. Leist and U. Kraushaar
Arch Toxicol. 2021 Jan;95(1):229-252. DOI: 10.1007/s00204-020-02956-3. PMID: 33269408

"Neurotoxicity and underlying cellular changes of 21 mitochondrial respiratory chain inhibitors."
Delp, J., A. Cediel-Ulloa, I. Suciu, P. Kranaster, B. M. van Vugt-Lussenburg, V. Munic Kos, W. van der Stel, G. Carta, S. H. Bennekou, P. Jennings, B. van de Water, A. Forsby and M. Leist
Arch Toxicol. 2021 Feb;95(2):591-615. DOI: 10.1007/s00204-020-02970-5.PMID: 33512557

"Neurodevelopmental toxicity assessment of flame retardants using a human DNT in vitro testing battery."
Klose, J., M. Pahl, K. Bartmann, F. Bendt, J. Blum, X. Dolde, N. Forster, A. K. Holzer, U. Hubenthal, H. E. Kessel, K. Koch, S. Masjosthusmann, S. Schneider, L. C. Sturzl, S. Woeste, A. Rossi, A. Covaci, M. Behl, M. Leist, J. Tigges and E. Fritsche
Cell Biol Toxicol. 2021 May 10. DOI: 10.1007/s10565-021-09603-2. PMID: 33969458

"Functional alterations by a subgroup of neonicotinoid pesticides in human dopaminergic neurons."
Loser, D., M. G. Hinojosa, J. Blum, J. Schaefer, M. Brüll, Y. Johansson, I. Suciu, K. Grillberger, T. Danker, C. Möller, I. Gardner, G. F. Ecker, S. H. Bennekou, A. Forsby, U. Kraushaar and M. Leist
Arch Toxicol. (b) 2021 Jun;95(6):2081-2107. DOI: 10.1007/s00204-021-03031-1. PMID: 33778899

"New approach methods (NAMs) supporting read-across: Two neurotoxicity AOP-based IATA case studies."
Van der Stel, W., G. Carta, J. Eakins, J. Delp, I. Suciu, A. Forsby, A. Cediel-Ulloa, K. Attoff, F. Troger, H. Kamp, I. Gardner, B. Zdrazil, M. J. Mone, G. F. Ecker, M. Pastor, J. C. Gomez-Tamayo, A. White, E. H. J. Danen, M. Leist, P. Walker, P. Jennings, S. Hougaard Bennekou and B. Van de Water
ALTEX. 2021;38(4):615-635. DOI: 10.14573/altex.2103051. PMID: 34114044

9.2 (Potential) linkage to AOPs

Test method could be potentially linked to the following AOPs in AOPwiki:

- AOP 3: Inhibition of the mitochondrial complex I of nigra-striatal neurons leads to parkinsonian motor deficits.
  → Organ effects: degeneration of DA neurons of nigrostriatal pathway

9.3 Steps towards mechanistic validation

a) LUHMES are dopaminergic, express DAT, TH. Are of human origin, form network
b) A formal mechanistic validation has not been performed.

c) The test covers a fundamental cellular process.

9.4 Pre-validation or validation

To date, 32 unique compounds (as defined by unique DTXSIDs) have been tested successfully in this assay (Delp 2019, Delp 2021).

No formal OECD 34 validation study has been done (eg., ring trials with a standard set of known positive and negative controls).

9.5 Linkage to (e.g. OECD) guidelines / regulatory use

Test is not linked to regulatory guidelines.

10. Test method transferability

10.1 Operator training

Experiences are required in:
- LUHMES cell culture
- multistep pipetting
- handling of XF Seahorse analyzer and its software
- Microsoft Excel
- GraphPad Prism
- Seahorse Analytics

Operator is trained and guided by a highly experienced instructor. Approximately 4 weeks will be needed for a smooth assay performance.

10.2 Transfer

Test system (UKN LUHMES cells) has been transferred and established to numerous other labs.

11. Safety, ethics and specific requirements

11.1 Specific hazards; issues of waste disposal

No specific requirements.

11.2 Safety data sheet (SDS)

SDS are available in the university DaMaRIS database (**Da**ngerous **Ma**terials **R**egistry **I**nformation **S**ystem).

11.3 Specific facilities / licenses

Work requires S1 cell culture laboratories (genetically modified cells).
No specific facilities are required.
No specific ethical approval is required.